

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403015
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Barrus; Michael et al.

Expandable implant with worm gear

Abstract

An expandable device comprising, a body defining a bore, a shaft received in the bore of the body, an end plate coupled to the shaft, wherein rotation of the shaft translates the end plate with respect to the body, and a locking mechanism engaged with the shaft so as to permit the shaft to rotate in a first direction and apply a resistance force to resist the shaft when attempting to rotate in a second direction.

Inventors:	Barrus; Michael (Redondo Beach, CA), Boyd; Clint (Leesburg, VA), Barnes; Liam Patrick (Leesburg, VA), Gantick; Jason Noel (Purcellville, VA), Wallenstein; Todd M. (Ashburn, VA), Carnes; Megan E. (Leesburg, VA)
Applicant:	K2M, Inc. (Leesburg, VA)
Family ID:	75690695
Assignee:	K2M, Inc. (Leesburg, VA)
Appl. No.:	17/917755
Filed (or PCT Filed):	April 07, 2021
PCT No.:	PCT/US2021/026189
PCT Pub. No.:	WO2021/207364
PCT Pub. Date:	October 14, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230165687 A1	Jun. 01, 2023

Related U.S. Application Data

us-provisional-application US 63006278 20200407

Publication Classification

Int. Cl.: A61F2/44 (20060101); A61F2/30 (20060101)

U.S. Cl.:

CPC A61F2/4455 (20130101); A61F2002/30367 (20130101); A61F2002/3052 (20130101); A61F2002/30525 (20130101); A61F2002/30579 (20130101); A61F2002/30784 (20130101)

Field of Classification Search

CPC: A61F (2/4455); A61F (2002/30367); A61F (2002/3052); A61F (2002/30525); A61F (2002/30579); A61F (2002/30784)

USPC: 623/17.11-

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8303663	12/2011	Jimenez et al.	N/A	N/A
8574300	12/2012	McManus	623/17.11	A61F 2/44
8636746	12/2013	Jimenez et al.	N/A	N/A
8992620	12/2014	Ashley et al.	N/A	N/A
10420654	12/2018	Logan et al.	N/A	N/A
2011/0015747	12/2010	McManus	623/17.16	A61F 2/44
2011/0160861	12/2010	Jimenez	623/17.16	A61F 2/4455
2012/0158071	12/2011	Jimenez	606/86A	A61F 2/447
2015/0094814	12/2014	Emerick	623/17.16	A61F 2/4455
2017/0119542	12/2016	Logan	N/A	A61F 2/4465

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2009064787	12/2008	WO	N/A
2013158294	12/2012	WO	N/A

OTHER PUBLICATIONS

International Search Report including Written Opinion for PCT/US2021/026189 issued Jul. 20, 2021; 12 pages. cited by applicant

Primary Examiner: Weiss; Jessica

Attorney, Agent or Firm: Lerner David LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/US2021/026189, filed on Apr. 7, 2021, which claims the benefit of the filing date of U.S. Provisional Application No. 63/006,278, filed Apr. 7, 2020, entitled Expandable Implant With Worm Gear, the disclosures of which are hereby incorporated herein by reference.

BACKGROUND

(1) Intervertebral fusion procedures involve removing at least a portion of a spinal disc between two vertebrae and implanting an intervertebral device between the two vertebrae. One example of an intervertebral device may include an expandable fusion device that can be inserted into an intervertebral space with a low profile and then expanded. Such a device can provide for a more minimally invasive surgical approach and may ease the insertion of the device between the vertebrae. Expandable fusion devices may also provide a distraction force between the two vertebrae. In this manner, the expandable fusion device can expand the vertebral foramen and prevent compression of the adjacent nerves, as well as aiding in fusing the adjacent vertebrae (such as by housing a bone growth material).

(2) However, such expandable devices may include the risk of over-expanding the vertebral disc and compressing other portions of the spine. Moreover, the expandable device may include the risk of collapsing after implantation, creating instability within the spine and leading to improper intervertebral fusion. Additionally, expanding the expandable device can require a great deal of force given the weight compressing on the intervertebral device small enough to fit between two vertebrae. Therefore, further improvements are desirable.

BRIEF SUMMARY

(3) In accordance with an aspect of the disclosure, an expandable device comprising, a body defining a bore, a shaft received in the bore of the body, an end plate coupled to the shaft, wherein rotation of the shaft translates the end plate with respect to the body, and a locking mechanism engaged with the shaft so as to permit the shaft to rotate in a first direction and apply a resistance force to resist the shaft when attempting to rotate in a second direction. At least one of the end plate and the body may include a textured surface. At least one of the end plate and the body may include a plurality of pores. The end plate may define an opening and the body defines an opening, the opening of the end plate being in communication with the opening of the body. The body may define a first opening transverse to the bore, the device further comprising a first gear sleeve threadably received in the first threaded opening of the body. The first gear sleeve may define a channel, the end plate including a first post received in the channel of the first gear sleeve. The first gear sleeve may include gear teeth, the shaft having a first threaded section engaged with the gear teeth. The expandable may further comprise a pin, wherein the body defines a hole configured to receive the pin. The shaft may define an indent configured to engage with the pin such that the worm gear is axially fixed to the body. The expandable device may further comprise a spring applying a spring force to the locking mechanism. The shaft may include a cap at an end of the shaft, the cap configured to engage the locking mechanism. The cap may be tapered with a first taper angle and the locking mechanism may be tapered with a second taper angle, the second taper angle being more acute than the first taper angle such that the locking mechanism applies the resistance force to the cap when the locking mechanism is engaged to the cap. The cap may have a first ratchet teeth and the locking mechanism may have a second ratchet teeth configured to engage the first ratchet teeth, the second ratchet teeth applying the resistance force to the first ratchet teeth. The bore may include a first section and a second section, the first section configured to receive the cap and the second section configured to receive the body, the first section having a first diameter

and the second section having a second diameter, the first diameter being larger than the second diameter. The locking mechanism may include an extension and the first section may include a first portion radially extending from the first section, the first portion configured to receive the extension such that the locking mechanism is rotationally fixed within the first section. The locking mechanism may include a pawl having a ratchet tooth configured to engage teeth on the worm gear. The expandable may further comprise a ball bearing configured to engage with the gear teeth of the first gear sleeve. The body may include a second opening, the expandable device may further comprise a second gear sleeve received in the second opening, the second gear sleeve having gear teeth and defining a channel therethrough, the end plate may have a second post received in the channel of the second gear sleeve, and the shaft may have a second threaded section engaged with the gear teeth of the second gear sleeve.

(4) In accordance with another aspect of the disclosure, an expandable device comprising, a body defining a bore and a first opening transverse to the bore, a shaft received in the bore of the body, a first gear sleeve received in the first opening and defining a channel, the first gear sleeve having a stopping structure within the channel, an end plate having a first post received in the channel of the first gear sleeve, and a stop plate engaged with the first post and received in the channel, wherein rotation of the shaft translates the end plate to a maximum distance determined by the engagement between the stop plate and the stopping structure. The post of the end plate may define a threaded opening and the stop plate may include a post, the post of the stop plate received in the threaded opening of the post of the end plate. The stop plate may define a hole radially distant from an axis defined by the post and configured to receive a set screw. The channel of the first gear sleeve may define a first threaded section configured to receive the post of the end plate and a second section configured to receive the stop plate. The stopping structure may lie between the first threaded section and the second section. At least one of the end plate and the body may include a textured surface. At least one of the end plate and the body may include a plurality of pores. The end plate may define an opening and the body may define an opening, the opening of the end plate being in communication with the opening of the body. The first gear sleeve may include gear teeth, the shaft having a first threaded section engaged with the gear teeth. The body may include a second opening, the expandable device may further comprise a second gear sleeve received in the second opening, the second gear sleeve having gear teeth and defining a channel therethrough, the end plate may have a second post received in the channel of the second gear sleeve, and the shaft may have a second threaded section engaged with the gear teeth of the second gear sleeve. The expandable may further comprise a pin, wherein the body may define a hole configured to receive the pin. The shaft may define an indent configured to engage with the pin such that the worm gear is axially fixed to the body.

(5) In accordance with another aspect of the disclosure, a method of using an expandable device having a body, comprising, rotating a shaft in a first rotational direction within the body to translate an end plate in a first axial direction, and locking the end plate in a first position along the first axial direction by engaging the shaft with a locking mechanism applying a resistance force to the shaft to resist the shaft from rotating in a second rotational direction opposite the first rotational direction. The method may further comprise, after locking the end plate, further rotating the worm gear in the second rotational direction by applying a rotation force greater than the resistance force. The shaft may include a taper having a first angle and the locking mechanism may have an opening having a taper with a second angle, the second angle being more acute than the first angle, wherein locking the end plate may include the resistance force being applied from the taper of the shaft being received in the opening of the locking mechanism. The method may further comprise unlocking the end plate by applying a pushing force to the shaft to separate the taper of the shaft and the opening of the locking mechanism. The shaft may include ratchet teeth and the locking mechanism may include ratchet teeth, wherein locking the end plate may include the resistance force being applied from the ratchet teeth of the locking mechanism engaging with the ratchet teeth

of the shaft. The method may further comprise unlocking the end plate by applying a pushing force to the locking mechanism to disengage the ratchet teeth of the locking mechanism and the ratchet teeth of the shaft. The expandable device may include a spring applying a spring force to the locking mechanism, wherein applying the pushing force may include the pushing force being greater than the spring force. The method may further comprise rotating the shaft in the second rotational direction so as to further translate the end plate in a second axial direction opposite the first axial direction. The method may further comprise locking the end plate in a second position by engaging the shaft with the locking mechanism applying the resistance force to the shaft. Rotating the shaft may include axially fixing the shaft with a pin received in an indent of the shaft while the shaft rotates. The method may further comprise inserting an actuating mechanism within an opening of the shaft. The method may further comprise inserting an inserter within a second threaded opening of the body. The method may further comprise inserting a bone growth material within at least one of an opening or pores within at least one of an end plate or body. The method may further comprise rotating a first gear sleeve within a first opening of the body and about a first post of the end plate, the first gear sleeve defining a channel threadably receiving the first post. The method may have a first threaded section and the first gear sleeve may have gear teeth engaged with the first threaded section, and wherein rotating the shaft further rotates the first gear sleeve due to engagement between the first threaded section of the shaft and the gear teeth of the first gear sleeve. The body may include a second opening, the expandable device further comprises a second gear sleeve threadably received in the second opening, the second gear sleeve having gear teeth and a defining a channel therethrough, the end plate may have a second post received in the channel of the second gear sleeve, the shaft may have a second threaded section engaged with the gear teeth of the second gear sleeve, and the method may further comprise, rotating the second gear sleeve within the second opening of the body and about the second post of the end plate.

(6) In accordance with another aspect of the disclosure, a method of using an expandable device having a body, comprising, rotating a shaft within the body in a first rotational direction about a longitudinal axis of the shaft so as to rotate a first gear sleeve about an axis extending in a first axial direction transverse to the longitudinal axis, the rotation of the first gear sleeve causing a post connected to an end plate to translate respect to the first gear sleeve along the first axial direction, the first gear sleeve defining a channel receiving the post of the end plate therein, and stopping the translation of the end plate by engaging a stop plate coupled to the post with a stopping structure of the first gear sleeve. The channel of the first gear sleeve may define a first threaded section and a second section, the stopping structure being in between the first threaded section and the second section, wherein the first post of the end plate is received through the first threaded section and the stop plate is received in the second section, wherein translating the end plate may include translating the post of the end plate within the first threaded section and translating the stop plate within the second section. Rotating the shaft may include axially fixing the shaft with a pin received in an indent of the shaft while the shaft rotates. The method may further comprise inserting an actuating mechanism within an opening of the shaft. The method may further comprise inserting an inserter within a second threaded opening of the body. The method may further comprise inserting a bone growth material within at least one of an opening or pores within at least one of an end plate or body. The shaft may have a first threaded section and the first gear sleeve may have gear teeth engaged with the first threaded section, and wherein rotating the shaft further rotates the first gear sleeve due to engagement between the first threaded section of the shaft and the gear teeth of the first gear sleeve. Rotating the shaft within the body in the first rotational direction may drive a second gear sleeve to rotate about a second axis extending in the first axial direction, the rotation of the second gear sleeve causing a second post connected to the end plate to translate along the first axial direction within a channel of the second gear sleeve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other features, aspects, and advantages of the present invention will become better understood with regard to the following description, and accompanying drawings.
- (2) FIG. 1 depicts an exploded view of an expandable device according to one embodiment of the disclosure.
- (3) FIG. 2A depicts a side view of an end plate of the expandable device of FIG. 1.
- (4) FIG. 2B depicts a top view of the end plate of FIG. 2A.
- (5) FIG. 2C depicts a cross-sectional view of the end plate of FIG. 2A along the E-E axis.
- (6) FIG. 3A depicts a first side view of a body of the expandable device of FIG. 1.
- (7) FIG. 3B depicts a second side view, opposite the first side view, of the body of FIG. 3A.
- (8) FIG. 3C depicts a top view of the body of FIG. 3A.
- (9) FIG. 3D depicts a cross-sectional view of the body of FIG. 3A along the A-A axis.
- (10) FIG. 3E depicts a rear view of the body of FIG. 3A.
- (11) FIG. 3F depicts a cross-sectional view of the body of FIG. 3A along the B-B axis.
- (12) FIG. 3G depicts a cross-sectional view of the body of FIG. 3A along the C-C axis.
- (13) FIG. 4A depicts an isometric view of a right-handed gear sleeve of the expandable device of FIG. 1.
- (14) FIG. 4B depicts a top view of the right-handed gear sleeve of FIG. 4A.
- (15) FIG. 4C depicts a bottom view of the right-handed gear sleeve of FIG. 4A.
- (16) FIG. 4D depicts a cross-sectional view of the gear sleeve of FIG. 4A, sectioned by a plane extending along the rotational axis of the gear sleeve.
- (17) FIG. 5A depicts an isometric view of a worm gear of the expandable device of FIG. 1.
- (18) FIG. 5B depicts a cross-sectional view of the worm gear of FIG. 5A along its longitudinal axis.
- (19) FIG. 6A depicts an isometric view of a locking mechanism of the expandable device of FIG. 1.
- (20) FIG. 6B depicts a top view of the locking mechanism of FIG. 6A.
- (21) FIG. 7A depicts an isometric view of a stop plate of the expandable device of FIG. 1.
- (22) FIG. 7B depicts a bottom view of the stop plate of FIG. 7A.
- (23) FIG. 7C depicts a cross-sectional view of the stop plate of FIG. 7A along the K-K axis.
- (24) FIG. 8A depicts a schematic isometric view of a portion of a locking mechanism of an expandable device according to another embodiment of the disclosure.
- (25) FIG. 8B depicts a cross-sectional view of a component of a locking portion of the expandable device of FIG. 8A.
- (26) FIG. 9 depicts an isometric view of internal components of an expandable device according to another embodiment of the disclosure.
- (27) FIG. 10 depicts an enlarged isometric view of a component of a locking mechanism of an expandable device according to another embodiment of the disclosure.
- (28) FIG. 11 depicts an enlarged isometric view of a component of a locking mechanism of an expandable device according to another embodiment of the disclosure.
- (29) FIG. 12 depicts an isometric view of a component of an expandable device according to another embodiment of the disclosure.
- (30) FIG. 13 depicts an isometric view of an expandable device according to another embodiment of the disclosure.
- (31) FIG. 14 depicts a top view of an expandable device according to another embodiment of the disclosure.
- (32) FIG. 15 depicts a cross-sectional elevation view of the expandable device of FIG. 14 in a collapsed condition.

(33) FIG. 16 depicts a cross-sectional view of the expandable device of FIG. 14 in an expanded condition.

DETAILED DESCRIPTION

(34) When referring to specific directions and planes in the following disclosure, it should be understood that, as used herein, the term “proximal” means closer to the operator/surgeon, and the term “distal” means further away from the operator/surgeon. The term “anterior” means toward the front of the body or the face, and the term “posterior” means toward the back of the body. With respect to the longitudinal axis of the spine, the term “superior” refers to the direction towards the head, and the term “inferior” refers to the direction towards the pelvis and feet. The “transverse plane” is that plane which is orthogonal to the longitudinal axis of the spine. The “coronal plane” is a plane that runs from side to side of the body along the longitudinal axis of the spine and divides the body into anterior and posterior portions. The “sagittal plane” is a plane that runs along the longitudinal axis of the spine and defines a plane of symmetry that separates the left and right sides of the body from each other. Finally, the “medial” refers to a position or orientation toward the sagittal plane, and lateral refers to a position or orientation relatively further from the sagittal plane.

(35) FIG. 1 depicts an exploded view of an expandable device 100. Device 100 expands toward an expanded condition when end plate 110 is further away from body 120 and collapses towards a collapsed condition when the end plate is closer to the body.

(36) As discussed further below, right-hand gear sleeve 140a and left-hand gear sleeve 140b threadably receive end plate 110 while body 120 threadably receives the gear sleeves. As a result, rotation of the gear sleeves within the body creates a telescoping expansion in which the gear sleeves translate relative to the body and the plate translates relative to the gear sleeves (and, in turn, the body), similar to that disclosed in U.S. Pat. No. 8,303,663 (“the '663 Patent”), the entire disclosure of which is hereby incorporated herein by reference. Worm gear (or shaft) 150 is received within body 120, being axially held in place by pin 190 received within the body, and is threadably engaged with gear sleeves 140a,b such that rotation of the worm gear rotates the gear sleeves. A maximum translating distance is set by contact between stop posts 170 and gear sleeves 140a,b. Locking mechanism 160 is received in body 120 such that the locking mechanism is rotationally fixed relative to worm gear 150. In this manner, the ratcheting engagement between locking mechanism 160 and worm gear 150 allows for the worm gear to rotate in one direction but not the other, thereby preventing back driving of the worm gear. Spring 180 applies a spring force to locking mechanism 160 and worm gear 150 such that locking mechanism 160 can passively maintain engagement with worm gear 150.

(37) FIGS. 2A-2C depicts end plate 110 having a proximal end 114 and a distal end 115. End plate 110 is configured to interface with an adjacent vertebra. Recess 112 is positioned adjacent to proximal end 114 and is configured to receive protrusion 122 of body 120 shown in FIGS. 3A-3E when device 100 is in a collapsed condition. Although distal end 115 is depicted as being inferiorly angled to allow for easier insertion into an intervertebral space, in alternative embodiments, end plate 110 may be substantially planar along its entire length.

(38) End plate 110 has a textured surface 111 with ridges shaped to better generate friction and prevent slippage with an adjacent vertebra. As shown in the figures, the ridges may be substantially triangular, and each ridge may be asymmetric on its proximal and distal sides. For example, the distal side of each ridge may be sloped gently with respect to the adjacent vertebral surface, while the proximal side of each ridge may be more sharply angled with respect to the vertebral surface. Such an arrangement may make it easier to insert the device 100 distally into the intervertebral space, while resisting proximal movement. In other embodiments, however, the ridges may be symmetrical on their proximal and distal sides, such as shown in FIGS. 1A-1C of the '663 Patent. Alternatively, end plate 100 may have a generally smooth surface, or alternative textured surfaces, such as an etched surface, a scored surface or a notched surface.

(39) End plate 110 includes pores 116 extending therein or therethrough to facilitate bone ingrowth.

In addition, or alternatively, opening **113** may be provided, which communicates with the interior of the device **100**, including opening **123** of body **120** shown in FIG. **3C**. In that way, bone graft material provided inside device **100** may communicate with the adjacent vertebral bodies to facilitate bone growth over time. Alternatively, end plate **100** may be solid with no openings **113** or pores **116**, or include only one of the opening or pores.

(40) Posts **117a,b** extend from end plate **110** on an opposite side of the end plate to textured surface **111**. Posts **117a,b** are configured to be respectively received in gear sleeves **140a,b**, as depicted in FIGS. **1** and **4A-4D**. Posts **117a,b** have an exterior threaded surface configured to engage with a threaded section of a channel defined within gear sleeves **140a,b** such that the gear sleeves are rotatably engaged with the posts. Thus, rotation of the gear sleeves causes the posts to telescopically translate in a superior or inferior direction with respect to the gear sleeves as the gear sleeves translate with respect to the body **120**. Further, posts **117a,b** of end plate **110** each have an internal threaded surface in threaded openings **118a,b**, as depicted in FIG. **2C**, configured to receive stop post **172** of stop plate **170** depicted in FIGS. **7A** and **7C** such that the contact between the stop plate and gear sleeves **140a,b** prevents end plate **110** from expanding too far. The external threads of post **117a** are left-hand threaded while the external threads of post **117b** are right-hand threaded. It is believed that such an arrangement can allow for a greater distraction force to be applied to the adjacent vertebra, as well as a decreased amount of friction between the posts and the gear sleeves when expanding and collapsing device **100**. However, in alternative embodiments, the external threading may be switched, or both posts **117a,b** may have the same threading direction.

(41) FIGS. **3A-3G** depict body **120** having a proximal end **124** and a distal end **125** with a textured surface **121** similar to textured surface **111** of end plate **110**. Body **120** includes threaded openings **128a,b** configured to receive posts **117a,b** of end plate **110**. Threaded openings **128a,b** define a channel and windows **138a,b** along a length of the channel. Windows **138a,b** allow for threaded openings **128a,b** to be in communication with second section **137** of bore **132**. In this manner, when worm gear **150**, shown in FIGS. **5A-5B**, is received in bore **132** and gear sleeves **140a,b**, shown in FIGS. **4A-4D**, are respectively received in threaded openings **128a,b**, threaded sections **157a,b** of the worm gear can engage with gear teeth **143a,b** of the gear sleeves. Although distal end **125** is depicted as being tapered in both superior-inferior and transverse planes, in alternative embodiments, the distal end of body **120** may have a substantially similar cross-sectional surface area as proximal end **124**.

(42) Body **120** has pores **126** and an opening **123** similar to pores **116** and opening **113** of end plate **110**. The opening **123** may define a channel extending through the interior of the body **120**. Body **120** additionally includes pores **129**, **130**, adjacent distal tip **125**, running along a medial and lateral direction through at least a portion of the body and serving a similar purpose as pores **126** and opening **123**. Pores **130** may further communicate with opening **123**, for example to allow bone graft material within the opening **123** to pass through the side of the body **120**. Body **120** further includes a pin hole **127** for receiving a pin **190**, as depicted in FIG. **1**. Alternative embodiments of body **120** may include more or fewer pores or openings, including no pores or openings.

(43) As shown in FIG. **3E**, body **120** includes a threaded opening **134** and bore **132**. Threaded opening **134** is located at proximal end **124** and configured to engage with an externally threaded portion of an inserter (not shown) such that the inserter may assist in delivering device **100** within the intervertebral space. Notch **131** is located adjacent bore threaded opening **134** and is configured to receive a protrusion (not shown) of the inserter to prevent rotation of device **100** when the device is engaged with the inserter. Bore **132** includes a first section **133** and second section **137**. Second section **137** is configured to receive a body **158** of worm gear **150** shown in FIGS. **5A-5B**, while first section **133** is configured to receive cap **151** of the worm gear. Although FIG. **3F** depicts first section **133** having a larger diameter than second section **137**, in alternative embodiments, the first section and the second can have the same diameter, or the second section can have a smaller diameter than the first section. First section is configured to receive spring **180**, locking mechanism

160 and cap **151** of worm gear **150**. First section **133** includes portions **135**, **136** radially extending from the first section. First portion **135** and second portion **136** of first section **133** are configured to respectively receive extension **163** and extended ratchet tooth **161** of locking mechanism **160** shown in FIG. **6A**. As discussed further below, the engagement between locking mechanism **160** and first section **133** allows axial movement of the locking mechanism but prevents rotation of the locking mechanism such that worm gear **150** can be rotated with respect to the locking mechanism.

(44) FIGS. **4A-4D** depicts right-hand gear sleeve **140a** having a superior end **141a** and an inferior end **142a**, and defining a channel **145a** therethrough. Right-hand gear sleeve **140a** and left-hand gear sleeve **140b** share the same features except the right-hand gear sleeve has opposite-handed external threads to the left-hand gear sleeve. Channel **145a** includes a threaded section **148a** configured to receive and threadably mate with post **117a** of end plate **110** while exterior threads **144a** are configured to be received in and mate with an interior thread of threaded openings **128a** of body. Exterior threads **144a** include gear teeth **143a** cut into the exterior threads and configured to engage with thread sections **157a** of worm gear **150** such that rotation of gear teeth **143a** by worm gear **150** rotates gear sleeve **140a** to translate in superior or inferior direction. Rotation of gear sleeve **140a** also rotates the gear sleeve about post **117a** of end plate **110** such that both the gear sleeve and end plate are translated relative to each other as they both translate away from body **120**, thereby creating a telescopic movement to expand or collapse device **100**. Although gear teeth **143a** are depicted as being cut down to the root or minor diameter of exterior threads **144a**, in alternative embodiments, the gear teeth may only be cut a distance between the major and minor diameter of the exterior thread to maximize the strength of both the exterior threads and gear teeth.

(45) As shown in FIG. **4D**, threaded section **148a** has a length defined between superior end **141a** and ledges (or stopping structures) **146**. Second portion **147a** is defined within the interior of right-hand gear sleeve **140a** between ledges **146** and inferior end **142a**. Second portion **147a** is configured to receive stop plate **170**. Ledges **146a** are configured to engage with a superior surface **175** of stop cap **171** when stop plate **170** is received within gear sleeve **140a** and post **117a**, as described further below.

(46) FIGS. **5A-5B** depicts worm gear **150** having cap **151** at a proximal end **154** and body **158** with a distal end **155**. The cap **151** may be monolithically formed with the body **158**, or the cap **151** may be separately formed and joined to the body **158**. As described above, cap **151** is configured to be received within first section **133** of bore **132** in body **120** while ratchet teeth **156** are configured to engage with extended ratchet tooth **161** and ratchet teeth **162** of locking mechanism **160**. This ratcheting mechanism allows worm gear **150** to be rotated in a first direction but prevented from rotating in an opposite direction relative to locking mechanism **160**. Cap **151** includes an opening **152** configured to be actuated by an actuation mechanism (not shown). For example, this can include a hex driver, screw driver, or the like. Worm gear **158** is axially fixed to body **120** through pin **190** being at least partially received within indent **153**.

(47) Body **158** includes thread sections **157a,b**. Thread section **157a** has right-hand threads configured to engage exterior threads **144a** of right-hand gear sleeve **140a** while thread section **157b** has left-hand threads configured to engage the corresponding exterior threads of left-hand gear sleeve **140b**. In this manner, rotation of cap **151** rotates body **158** which, in turn, rotates gear sleeves **140a,b**, translating both the gear sleeves and end plate **110** to expand or collapse device **100**. The opposite-hand engagement between thread sections **157a,b** and the exterior threads of gear sleeves **140a,b** is believed to decrease the amount of friction between worm gear **150** and the gear sleeves while allowing for a greater distraction force to be applied to the adjacent vertebra when expanding and collapsing device **100**.

(48) FIGS. **6A-6B** depicts locking mechanism **160** having ratchet teeth **162** and extension **163** defined between a proximal end **164** and a distal end **165**. The proximal end **164** of extension **163** may serve as an interface for a mechanism (e.g., a pin) on the inserter to push the locking mechanism **160** distally to disengage the ratcheting engagement and allow the device to collapse,

as discussed below. Extended ratchet tooth **161** is similar in structure to any teeth of ratchet teeth **162** except the extended ratchet tooth is thicker than the other ratchet teeth. Specifically, an extended portion protrudes radially outwardly from an exterior surface of extended ratchet tooth **161**. As described above, locking mechanism **160** is configured to be received within first section **133** of bore **132** such that extension **163** is received within first portion **135** and extended ratchet tooth **161** is received within second portion **136**. In this manner, once locking mechanism **160** and worm gear **150** are received within body **120**, locking mechanism is rotationally fixed such that the worm gear can rotate relative to the locking mechanism.

(49) Moreover, extended ratcheting tooth **161** and ratchet teeth **162** of locking mechanism **160** are configured to engage with ratchet teeth **156** of worm gear **150** in a ratchet-like manner. As such, rotation of worm gear **150** in a first, clockwise direction would allow the sloped portions of ratchet teeth **156** of the worm gear to slide along the sloped portions of ratcheting tooth **161** and ratchet teeth **162** of locking mechanism **160** while the ratchet teeth of the locking mechanism applies a resistance force to the ratchet teeth of the worm gear to prevent the worm gear from rotating in the opposite, counter-clockwise direction. In this manner, after rotation of worm gear **150** in a first direction to rotate gear sleeves **140a,b** to expand end plate **100**, the position of end plate **100** can be locked in place through the engagement between locking mechanism **160** and cap **151** of the worm gear. In this manner, once device **100** has been expanded to a desired position, the engagement between locking mechanism **160** and worm gear **150** prevents the worm gear from back driving and unintentionally collapsing the device.

(50) FIGS. 7A-7C depict stop plate **170** having stop cap **171** and stop post **172**. Stop cap **171** may be welded to stop post **172** through holes **173**. Stop post **172** extends from stop cap **171** and is configured to be received in threaded openings **118a,b** when the stop cap and the posts **117a,b** of end plate **110** are received in gear sleeves **140a,b**. In this manner, the engagement between superior surface **175** of stop cap **171** and ledges **146** of gear sleeves **140a,b** can set a limit on the distance traveled by end plate **110** while device **100** is in its expanded condition.

(51) In an alternative embodiment, the locking mechanism may additionally or alternatively include a pawl to interact with ratchet teeth along the body of the worm gear. FIGS. 8A-8B, depicts a schematic representation of expandable device **200** including each of the features of device **100** except as discussed below. Worm gear **250** includes toothed section **259** located along a central portion of body **258** of the worm gear. Locking mechanism **260** is coupled to body **220** and includes pawl **263** having ratchet tooth **262**. The sloped portions of ratchet tooth **262** of locking mechanism **260** engage with the sloped portions of toothed section **259** such that worm gear **250** can be rotated in a first direction, such as a clockwise direction, but the axial portions of the ratchet teeth apply a resistance to the toothed section to prevent the worm gear from rotating in a second, opposite direction. Ratchet tooth **262** can be disengaged from toothed section **259** by either translating or pivoting the pawl **263** of locking mechanism **260** away from worm gear **250**. This ratcheting engagement assists in preventing back driving of worm gear **250** and unintentional collapsing of device **200**. Although not shown in FIG. 8A, worm gear **250** can include threaded sections, such as threaded sections **157a,b** shown in FIGS. 5A-5B, located on either side of toothed section **259**. In a further alternative, there can be multiple toothed sections **259** and multiple corresponding pawls **263**. In a yet further alternative, toothed section **259** is not located along a central portion of body **258** but rather can be placed along any portion of the body.

(52) In a further alternative embodiment, the locking mechanism may instead take the form of a taper lock. For example, FIG. 9 depicts expandable device **300** including each of the features of device **100**, **200** except as discussed below. Worm gear **350** includes tapered cap **351** and a threaded body **358** configured to engage with gear sleeves **340a,b**. The gear sleeves **340a,b** in this embodiment would both be threaded in the same way (i.e., either both right-handed or both left-handed), and the thread sections of the worm gear **350** would also be threaded in the same way as one another, or the body **358** could be entirely threaded, as shown in FIG. 9. Once worm gear **350**

is inserted into the bore within body **320** and engaged with gear sleeves **340a,b**, locking mechanism **360** is inserted over cap **351** to retain the worm gear **350** within the bore. Opening **334** of locking mechanism **360** is shaped to have a more acute taper than the taper of cap **351**, however the opening is proximally offset from the cap such that the cap would have to travel a proximal distance in order to mate with the opening. In this manner, once worm gear **350** has been rotated to expand device **100** to a desired height, downward pressure on the expanded end plate (not shown) would tend to cause the gear sleeves **340a,b** to rotate backwards towards an unexpanded position. Moreover, since both gear sleeves are threaded in the same direction, the backwards rotation would result in the teeth of the gear sleeves acting like pinions tending to push the worm gear **350** like a rack back in the proximal direction. In an alternative embodiment, a compression spring may be located at the distal end of the worm gear **350** to bias the worm gear in the proximal direction. As a result, frictional force will develop between cap **351** and tapered opening **334** that will prevent back driving of the worm gear **350**. To further expand or collapse device **100**, an actuation mechanism (not shown) is inserted within cap **351** and a slight distal force is applied to disengage cap **351** and opening **334** such that worm gear **350** is freely rotatable.

(53) In a yet further alternative embodiment, the locking mechanism can be a clutch plate interacting with a spring. For example, FIG. **10** depicts expandable device **400** including each of the features of device **100**, **200**, **300** except as discussed below. Spring **480** includes a tooth **481** engaged with ratchet teeth **456** connected to the worm gear **450** such that a spring force is applied to the ratchet teeth. In this manner, spring **480** acts as a passive locking mechanism for worm gear **450**, as rotation of the worm gear requires overcoming the force of spring **480** to turn the worm gear. Ratchet teeth **456** may include two sloped portions rather than one sloped portion and one axial portion (as in ratchet teeth **162** of locking mechanism **160** shown in FIGS. **6A-6B**). Thus, worm gear **450** may be rotated in either direction by overcoming the locking force applied by the spring **480**.

(54) Alternatively, rather than the spring having its own tooth for directly interacting with the ratchet teeth of the worm gear, the locking mechanism can include a lock washer biased by the spring into engagement with the ratchet teeth of the worm gear, where the lock washer includes one or more corresponding teeth for engaging the ratchet teeth. For example, FIG. **11** depicts expandable device **500** including each of the features of device **100**, **200**, **300**, **400** except as discussed below. Locking mechanism **560**, in the form of a lock washer, includes ratchet teeth **562** shaped to engage the ratchet teeth **556** of worm gear **550**. Like ratchet teeth **456** of FIG. **10**, ratchet teeth **556** and **562** of FIG. **11** may include two sloped portions rather than one sloped portion and one axial portion, so as to permit worm gear **550** to be rotated in either direction by overcoming the locking force of the spring **580** pressing ratchet teeth **562** of the locking mechanism **560** against ratchet teeth **556** of the worm gear **550**.

(55) In further alternatives (not shown), the ratchet teeth of the locking mechanisms of FIGS. **10** and **11** may include one sloped portion and one axial portion, so that worm gear may only be rotated in one direction without disengaging the locking mechanism.

(56) In an alternative embodiment, overexpansion of the end plate can be prevented by an interference object associated with one or more of the gear sleeves. For example, FIG. **12** depicts expandable device **600** including each of the features of device **100**, **200**, **300**, **400**, **500** except, rather than stop plate **170** shown in FIG. **1** setting a limit on expansion of device **100**, an interference object in the form of a ball bearing **690** can be positioned in gear teeth **643a** to stop gear sleeve **640** from further rotation when that object contacts the engaging threads of the worm gear.

(57) In a further alternative embodiment, the end plate and body can have an alternative shape. FIG. **13** depicts expandable device **700** including each of the features of device **100**, **200**, **300**, **400**, **500**, **600** except end plate **710** and body **720** both have a superior and inferior surface sloping outwardly from the proximal end towards the distal end. Such sloping surfaces may help with

lordotic correction of the adjacent vertebral bodies.

(58) In a yet further alternative embodiment, FIGS. **14-16** depict expandable device **800**, which may include any of the features of device **100**, **200**, **300**, **400**, **500**, **600**, and/or **700**, except that proximal end **814** of end plate **810** and body **820** may be angled obliquely relative to the longitudinal dimension of the device **800**. In this manner, device **800** may be inserted into a patient with proximal end **814** offset from and angled with respect to the longitudinal axis of the inserter. In that manner, device **800** may be better suited in a transforaminal lumbar interbody fusion (TLIF) procedure, whereas the straighter embodiments depicted above may be more suitable in a posterior lumbar interbody fusion (PLIF) procedure or a lateral approach, for example. In alternative embodiments, the angled surface along proximal end **814** of expandable device **80** may be angled in an opposite direction to that displayed in FIG. **14**.

(59) Although the above embodiments involve the end plate having two posts, in other embodiments, the end plate may have more or fewer than two posts. Further, the number of gear sleeves and thread sections on the worm gear may correspond to the number of posts (e.g., three gear sleeves and three threaded sections on the worm gear where the end plate has three posts).

(60) The expandable device may be constructed, in whole or in part, from one or more biocompatible metals, such as but not limited to any one of or any combination of titanium and its alloys, stainless steel and its alloys, magnesium and its alloys, cobalt and its alloys including a cobalt chrome alloy, nickel and its alloys, silver, tantalum, and niobium. It is also contemplated that any part of the expandable device may be made from one or more biocompatible polymers, such as but not limited to any one of or any combination of polyethylene (PE) and variations thereof, polyetheretherketone (PEEK), polyetherketone (PEK), acrylonitrile butadiene styrene (ABS), silicone, and cross-linked polymers.

(61) With reference to FIGS. **14-15**, methods of using expandable device **800** will now be described, however expandable devices **100**, **200**, **300**, **400**, **500**, **600**, **700** may be operated via similar methods, except as described below. As shown in FIG. **14**, device **800** is depicted in its collapsed condition and ready to be implanted within an intervertebral space between two adjacent vertebrae. An inserter (not shown) may first be engaged with threaded opening **834** to assist in manipulation and insertion of device **800** within the intervertebral space, leading with distal end **815**. The inserter may be disengaged after implanting device **800** or may stay engaged.

(62) Once device **800** has been implanted, a first actuation mechanism (not shown) can be inserted into opening **852** of worm gear **850** such that rotation of the first actuation mechanism in a first direction (e.g., a clockwise direction) correspondingly rotates the worm gear within bore **832** of body **820** while the longitudinal position of the worm gear **850** is maintained by pin **890** received in indent **853** of the worm gear. Rotation of worm gear **850** within bore **832** rotates the sloped portions of ratchet teeth **856** against the sloped portions of extended ratchet tooth **861** and ratchet teeth **862** of locking mechanism **860**. Additionally, rotation of worm gear **850** rotates gear sleeves **840a,b** through the engagement between gear teeth **843a,b** and thread sections **857a,b**. Rotation of worm gear **850** includes overcoming a slight resistance from spring **880** pressing locking mechanism **860** against cap **851** of the worm gear.

(63) Gear sleeves **840a,b** are rotated within threaded openings **828a,b** of body **820** through the engagement between exterior threads **844a,b** of the gear sleeves and the interior threads of the threaded openings of the body. Rotation of gear sleeves **840a,b** within threaded openings **828a,b** additionally translates the gear sleeves in a superior direction. Further, rotation of gear sleeves **840a,b** causes the gear sleeves to rotate about posts **817a,b** such that the engagement between threaded section **845a,b** of channel **848a,b** of the gear sleeves and the exterior threads of the posts translates the posts in a superior direction. As such, rotation of worm gear **850** simultaneously rotates gear sleeves **840a,b** about posts **817a,b** while also translating both the gear sleeves and the posts in a superior direction.

(64) Translation of posts **817a,b** is limited by the contact between superior surface **875** of stop plate

870 and ledges **846a,b** within second portion **847a,b** of gear sleeves **840a,b**. In this manner, rotation of worm gear **850** translates both gear sleeves **840a,b** and end plate **810** in a superior direction until a desired height, or where a maximum height is reached through the engagement between stop plate **870** and the gear sleeves.

(65) Once device **800** has finished expanding, locking mechanism **860** locks the position of worm gear **850** and end plate **810** in place. That is, locking mechanism **860** cannot rotate towards the unexpanded position due to the engagement between extension **863** and extended ratchet tooth **861** of locking mechanism **860** with the respective first portion **835** and second portion **836** of first section **833** of bore **832**. Therefore, worm gear **850** is prevented from rotating in a second counter-clockwise direction through the engagement and resistance force of the axial portions of extended ratchet tooth **861** and ratchet teeth **862** of the locking mechanism against the axial portions of ratchet teeth **856** of the worm gear. End plate **810** cannot translate without counter-clockwise rotation of worm gear **850** and corresponding rotation of gear sleeves **840a,b**, so locking mechanism **860** prevents the worm gear from rotating in the counter-clockwise direction, thereby locking the position of end plate **810** and preventing unintentional collapse of device **800**. In this manner, device **800** can maintain an expanded condition when a desired height of end plate **810** is reached without a separate step of locking the position of the end plate (e.g., through the use of a set screw, or the like). This increases the efficiency of the surgery and decreases the chance of complications arising.

(66) To transition device **800** from the expanded condition to the collapsed condition, a second actuation mechanism (not shown), such as a pin provided by the inserter, can enter first or second portions **835**, **836** to distally push extension **863** or extended ratchet tooth **861** of locking mechanism **860** distally into body **820**. That second actuation mechanism must overcome the force of spring **880** until extended ratchet tooth **861** and ratchet teeth **862** of the locking mechanism disengage from ratchet teeth **856** of worm gear **850**. In a preferred embodiment, the actuating mechanism engages the proximal end **864** of extension **863** of locking mechanism **860** to disengage ratchet teeth **856** of worm gear **850** from ratchet teeth **862** of locking mechanism **860**. While maintaining the second actuation mechanism in this position, such that locking mechanism **860** and worm gear **850** do not re-engage, the first actuation mechanism may rotate the worm gear counter-clockwise, correspondingly rotating gear sleeves **840a,b** within body **820** and posts **817a,b** to inferiorly translate the gear sleeves and end plate **810** until a desired height is reached (e.g., a collapsed condition).

(67) With reference to FIGS. **8A-8B**, device **200** may transition from the expanded condition to the collapsed condition by at least one of translating or pivoting pawl **263** of locking mechanism **260** away from worm gear **250** to disengage ratchet tooth **262** of the locking mechanism **260** from ratchet teeth **259** of the worm gear. Once locking mechanism **260** and worm gear **250** are disengaged, the worm gear can be rotated in a counter-clockwise direction to collapse device **200**.

(68) With reference to FIGS. **9A-9B**, the position of device **300** and worm gear **350** may be locked through the engagement between tapered cap **351** and opening **334** of locking mechanism **360**.

Opening **334** is configured to have a taper angle that is more acute than a taper angle of cap **351** such that, once device **300** is at a desired expanded condition, excess back driving of worm gear **350** would be prevented from the resistance, frictional force of the cap being mated with the opening. Worm gear **350** can be further rotated to further expand or collapse device **300** by an actuation mechanism (not shown) applying a slight distal force on cap **351** to disengage cap **351** from opening **334**.

(69) With reference to FIG. **10**, spring **480** passively locks the position of worm gear **450** through a spring force being applied to the engagement between tooth **481** and ratchet teeth **456** of the worm gear. In this manner, further rotation of worm gear **450** requires a rotational force to be applied to cap **451** greater than the spring force from spring **480**. Similarly, with reference to FIG. **11**, spring **580** passively locks the position of worm gear **550** through a spring force being applied to the

engagement between ratchet teeth **562** of locking mechanism **560** and ratchet teeth **556** of the worm gear.

(70) With reference to FIG. **12**, overexpansion of device **600** can be prevented through the use of ball bearing **690** being positioned within gear teeth **643a** to prevent further rotation of gear sleeve **640** and, in turn, further superior translation of the end plate (not shown).

(71) Although the disclosure herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present disclosure. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present disclosure as defined by the appended claims.

Claims

1. An expandable device comprising: a body defining a bore; a shaft received in the bore of the body; an end plate coupled to the body, wherein rotation of the shaft translates the end plate with respect to the body; and a locking mechanism engaged with the shaft so as to permit the shaft to rotate in a first direction and apply a resistance force to resist the shaft when attempting to rotate in a second direction, wherein the shaft includes a cap at an end of a body of the shaft, the cap configured to engage the locking mechanism, and wherein the cap has a first set of ratchet teeth and the locking mechanism has a second set of ratchet teeth configured to engage the first set of ratchet teeth, the second set of ratchet teeth applying the resistance force to the first set of ratchet teeth.
2. The expandable device of claim 1, wherein at least one of the end plate and the body includes a textured surface.
3. The expandable device of claim 1, wherein at least one of the end plate and the body includes a plurality of pores.
4. The expandable device of claim 1, wherein the end plate defines an opening and the body defines an opening, the opening of the end plate being in communication with the opening of the body.
5. The expandable device of claim 1, wherein the body defines a first opening transverse to the bore, the device further comprising a first gear sleeve threadably received in the first opening of the body.
6. The expandable device of claim 5, wherein the first gear sleeve defines a channel, the end plate including a first post received in the channel of the first gear sleeve, wherein the first gear sleeve includes gear teeth, the shaft having a first threaded section engaged with the gear teeth.
7. The expandable device of claim 6, wherein: the body includes a second opening; the expandable device further comprises a second gear sleeve received in the second opening, the second gear sleeve having gear teeth and defining a channel therethrough; the end plate has a second post received in the channel of the second gear sleeve; and the shaft has a second threaded section engaged with the gear teeth of the second gear sleeve.
8. The expandable device of claim 5, further comprising a ball bearing configured to engage with gear teeth of the first gear sleeve.
9. The expandable device of claim 1, further comprising a pin, wherein the body defines a hole configured to receive the pin, wherein the shaft defines an indent configured to engage with the pin such that the shaft is axially fixed to the body.
10. The expandable device of claim 1, further comprising a spring applying a spring force to the locking mechanism.
11. The expandable device of claim 1, wherein the cap is tapered with a first taper angle and the locking mechanism is tapered with a second taper angle, the second taper angle being more acute than the first taper angle such that the locking mechanism applies the resistance force to the cap when the locking mechanism is engaged to the cap.

12. The expandable device of claim 1, wherein the bore includes a first section and a second section, the first section configured to receive the cap and the second section configured to receive the body of the shaft, the first section having a first diameter and the second section having a second diameter, the first diameter being larger than the second diameter, wherein the locking mechanism includes an extension and the first section includes a first portion radially extending from the first section, the first portion configured to receive the extension such that the locking mechanism is rotationally fixed within the first section.
 13. The expandable device of claim 1, wherein the locking mechanism includes a pawl having a ratchet tooth configured to engage teeth on the shaft.
 14. An expandable device comprising: a body defining a bore and a first opening transverse to the bore; a shaft received in the bore of the body; a first gear sleeve received in the first opening and defining a channel, the first gear sleeve having a stopping surface within the channel; an end plate having a first post received in the channel of the first gear sleeve; and a stop plate engaged with the first post and received in the channel, wherein rotation of the shaft translates the end plate from a first position in which the stop plate is spaced from the stopping surface to a second position in which the stop plate abuts the stopping surface, thereby defining a limit of translation of the end plate.
 15. The expandable device of claim 14, wherein the first post of the end plate defines a threaded opening and the stop plate includes a post, the post of the stop plate received in the threaded opening of the first post of the end plate.
 16. The expandable device of claim 14, wherein the stop plate defines a hole radially distant from an axis defined by the first post and configured to receive a set screw.
 17. The expandable device of claim 14, wherein the channel of the first gear sleeve defines a first threaded section configured to receive the post of the end plate and a second section configured to receive the stop plate.
 18. The expandable device of claim 17, wherein the stopping structure lies between the first threaded section and the second section.
 19. The expandable device of claim 14, wherein the first gear sleeve includes gear teeth, the shaft having a first threaded section engaged with the gear teeth.
 20. The expandable device of claim 14, further comprising a pin, wherein the body defines a hole configured to receive the pin, wherein the shaft defines an indent configured to engage with the pin such that the shaft is axially fixed to the body.
-